

Carbon free energy for Google Cloud regions

In choosing which Google Cloud region to host your application, there are multiple considerations:

- Latency to your end users can be different from one region to the next.
- The price of services differs from region to region.
- The electricity used to power your application might have a different carbon intensity.

This document explains how to include carbon emissions characteristics into the location choice for your Google Cloud services.

A carbon-free cloud for our customers

To power each Google Cloud region, we use electricity from the grid where the region is located. This electricity generates more or less carbon emissions (gCO₂eq), depending on the type of power plants generating electricity for that grid and when we consume it. We set a goal (<https://cloud.google.com/blog/topics/inside-google-cloud/announcing-round-the-clock-clean-energy-for-cloud>)

to match our energy consumption with carbon-free energy (CFE), every hour and in every region by 2030.

As we work towards our 2030 goal, we want to empower our customers to leverage our 24/7 carbon free energy efforts and consider the carbon impact of where they locate their applications. To characterize each region we use a metric: "CFE%". This metric is calculated for every hour and tells us what percentage of the energy we consumed during an hour that is carbon-free, based on two elements:

1. The generation feeding the grid at that time (which power plants are running)
2. Google-attributed (<https://sustainability.google/progress/projects/ppa/>) clean energy produced onto that grid during that time.

We aggregate the available average hourly CFE percentage for each Google Cloud region for the year and have provided the 2024 data below.

Understanding the data

Google CFE%: This is the average percentage of carbon free energy consumed in a particular location on an hourly basis, while taking into account the investments we have made in carbon-free energy in that location. This means that in addition to the carbon free energy that's already supplied by the grid, we have added carbon-free energy generation in that location to reach our 24/7 carbon free energy objective

(<https://www.gstatic.com/gumdrop/sustainability/247-carbon-free-energy.pdf>). As a customer, this represents the average percentage of time your application will be running on carbon-free energy.

Grid carbon intensity (gCO₂eq/kWh): This metric indicates the average operational gross emissions per unit of energy from the grid. This metric should be used to compare the regions in terms of carbon intensity of their electricity from the local grid. For regions that are similar in CFE%, this will indicate the relative emissions for when your workload is not running on carbon free energy.

Carbon data across Google Cloud regions

Google Cloud Region	Location	Google CFE%	Grid carbon intensity (gCO ₂ eq/kWh)
africa-south1	Johannesburg	15%	657
asia-east1	Taiwan	17%	439
asia-east2	Hong Kong	1%	505
asia-northeast1	Tokyo	17%	453
asia-northeast2	Osaka	46%	296
asia-northeast3	Seoul	37%	357
asia-south1	Mumbai	9%	679
asia-south2	Delhi	29%	532
asia-southeast1	Singapore	4%	367

Google Cloud Region	Location	Google CFE%	Grid carbon intensity (gCO ₂ eq/kWh)	
asia-southeast2	Jakarta	18%	561	
australia-southeast1	Sydney	34%	498	
australia-southeast2	Melbourne	39%	454	
europa-central2	Warsaw	40%	643	
europa-north1	Finland	98%	39	 <u>Low CO₂</u> (https://cloud.google.com/sustainability/region-carbon#region-picker)
europa-north2	Stockholm	100%	3	 <u>Low CO₂</u> (https://cloud.google.com/sustainability/region-carbon#region-picker)
europa-southwest1	Madrid	87%	89	 <u>Low CO₂</u> (https://cloud.google.com/sustainability/region-carbon#region-picker)
europa-west1	Belgium	84%	103	 <u>Low CO₂</u> (https://cloud.google.com/sustainability/region-carbon#region-picker)
europa-west2	London	79%	106	 <u>Low CO₂</u> (https://cloud.google.com/sustainability/region-carbon#region-picker)
europa-west3	Frankfurt	68%	276	
europa-west4	Eemshaven	83%	209	 <u>Low CO₂</u> (https://cloud.google.com/sustainability/region-carbon#region-picker)
europa-west6	Zürich	98%	15	

Google Cloud Region	Location	Google CFE%	Grid carbon intensity (gCO ₂ eq/kWh)	
				<u>Low CO₂</u> (https://cloud.google.com/sustainability/region-carbon#region-picker)
europe-west8	Milan	73%	202	
europe-west9	Paris	96%	16	 <u>Low CO₂</u> (https://cloud.google.com/sustainability/region-carbon#region-picker)
europe-west10	Berlin	68%	276	
europe-west12	Turin	73%	202	
me-central1	Doha	1%	366	
me-central2	Dammam	1%	382	
me-west1	Tel Aviv	7%	434	
northamerica-northeast1	Montréal	99%	5	 <u>Low CO₂</u> (https://cloud.google.com/sustainability/region-carbon#region-picker)
northamerica-northeast2	Toronto	84%	59	 <u>Low CO₂</u> (https://cloud.google.com/sustainability/region-carbon#region-picker)
northamerica-south1	Mexico	19%	305	
southamerica-east1	São Paulo	88%	67	 <u>Low CO₂</u> (https://cloud.google.com/sustainability/region-carbon#region-picker)
southamerica-west1	Santiago	92%	238	 <u>Low CO₂</u> (https://cloud.google.com/sustainability/region-carbon#region-picker)

Google Cloud Region	Location	Google CFE%	Grid carbon intensity (gCO ₂ eq/kWh)	
us-central1	Iowa	87%	413	 <u>Low CO₂</u> (https://cloud.google.com/sustainability/region-carbon#region-picker)
us-central2	Iowa	88%	372	 <u>Low CO₂</u> (https://cloud.google.com/sustainability/region-carbon#region-picker)
us-east1	South Carolina	31%	576	
us-east2	Georgia	42%	340	
us-east4	Northern Virginia	62%	323	
us-east5	Columbus	62%	323	
us-south1	Dallas	94%	303	 <u>Low CO₂</u> (https://cloud.google.com/sustainability/region-carbon#region-picker)
us-west1	Oregon	87%	79	 <u>Low CO₂</u> (https://cloud.google.com/sustainability/region-carbon#region-picker)
us-west2	Los Angeles	63%	169	
us-west3	Salt Lake City	33%	555	
us-west4	Las Vegas	64%	357	

Find the same data in a machine readable format [on GitHub](https://github.com/GoogleCloudPlatform/region-carbon-info)

(<https://github.com/GoogleCloudPlatform/region-carbon-info>) or [as a BigQuery public dataset](https://console.cloud.google.com/marketplace/product/bigquery-public-datasets/regional-cfe) (<https://console.cloud.google.com/marketplace/product/bigquery-public-datasets/regional-cfe>).

The hourly grid mix and carbon intensity data used to calculate these metrics is from [Electricity Maps](https://www.electricitymap.org/map) (<https://www.electricitymap.org/map>). This data has not been assured.

How to incorporate carbon free energy in your location strategy

Be sure to consider the other [best practices for choosing resource locations](/solutions/best-practices-compute-engine-region-selection) (/solutions/best-practices-compute-engine-region-selection) like data residency requirements, latency to your end users, redundancy of the application, and price of the services available.

To use the CFE data above, here are some good ideas to get you started:

1. **Pick a cleaner region for your new applications.** If you are going to run an application over time, running in the region with the highest CFE% will emit the lowest carbon emissions.
2. **Run batch jobs on the cleanest option.** Batch workloads often have the benefit of planning. You should pick the region with the highest CFE% available to you.
3. **Set an organizational policy for low carbon locations.** You can restrict the location of your resources to a particular Google Cloud region or subset of regions using the "Resource Location Restriction" [organization policy](/resource-manager/docs/organization-policy/defining-locations) (/resource-manager/docs/organization-policy/defining-locations). Dedicated "[low carbon](/resource-manager/docs/organization-policy/defining-locations#value_groups)" [value groups](/resource-manager/docs/organization-policy/defining-locations#value_groups) (/resource-manager/docs/organization-policy/defining-locations#value_groups) have been created to enable you to restrict locations with [low carbon impact](#) (#region-picker). For example, if you want to use only US-based regions, use the "Low carbon United States" (`in:us-low-carbon-locations`) value group.

Low carbon indicators

Some location pages on the Google Cloud website and location selectors in the API Console display "



[Low CO₂](https://cloud.google.com/sustainability/region-carbon#region-picker) (https://cloud.google.com/sustainability/region-carbon#region-picker)" next to locations that have the lowest carbon impact. The "[Resource Location Restriction](/resource-manager/docs/organization-policy/defining-locations)" [organization policy](/resource-manager/docs/organization-policy/defining-locations) (/resource-manager/docs/organization-policy/defining-locations) offers "[low carbon](/resource-manager/docs/organization-policy/defining-locations#value_groups)" [value groups](/resource-manager/docs/organization-policy/defining-locations#value_groups) (/resource-manager/docs/organization-policy/defining-locations#value_groups).

For a location to be considered "low carbon", it must belong to a region with a **Google CFE% of at least 75%**, or, if CFE% information is not available, a grid carbon intensity of **maximum 200 gCO₂eq/kWh**.

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